The Approach

Requirements

Functional

- users can create & update an account
- users can login
- Users can manage their profiles.
- Employers can Create job postings.
- System can recommend jobs to candidates.
- System can recommend candidates to employers.
- matching service for candidates &jobs
- Candidates can Search for jobs.
- Employers can Search for candidates.
- System can Send notifications.
- System can Analyze skill gaps.
- Candidates can Browse and apply for jobs.
- Employers can manage application pipelines.(reject or interview)
- System Prevent duplicate applications

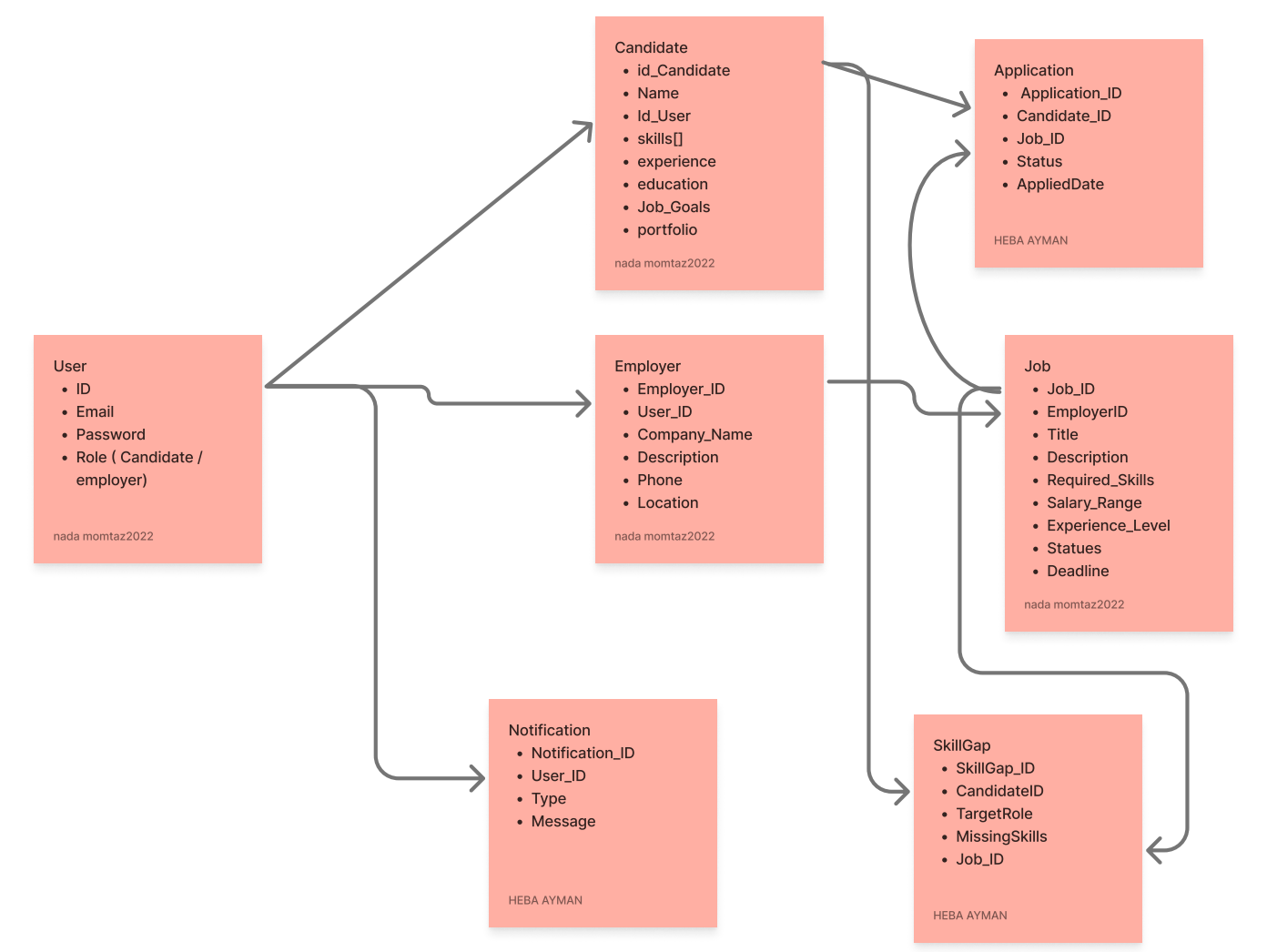
Non-Functional

- Scalability
- security
- Consistency
- low latency

Core Entities

- Candidates
- Employer
- Job
- Skill
- Application
- Notification

Data Model


User
ID
Email
Password
Role (Candidate / employer)
nada momtaz2022
Candidate
id_Candidate
Name
id_User
skills[]
experience
education
Job_Goals
portfolio
nada momtaz2022
Employer
Employer_ID
User_ID
Company_Name
Description
Phone
Location
nada momtaz2022
Application
Application_ID
Candidate_ID
Job_ID
Status
AppliedDate
HEBA AYMAN
Job
Job_ID
EmployerID
Title
Description
Required_Skills
Salary_Range
Experience_Level
Statues
Deadline
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Notification
Notification_ID
User_ID
Type
Message
HEBA AYMAN
SkillGap
SkillGap_ID
CandidateID
TargetRole
MissingSkills
Job_ID
HEBA AYMAN

Requirements

- POST /User/ register ——→ user-Id

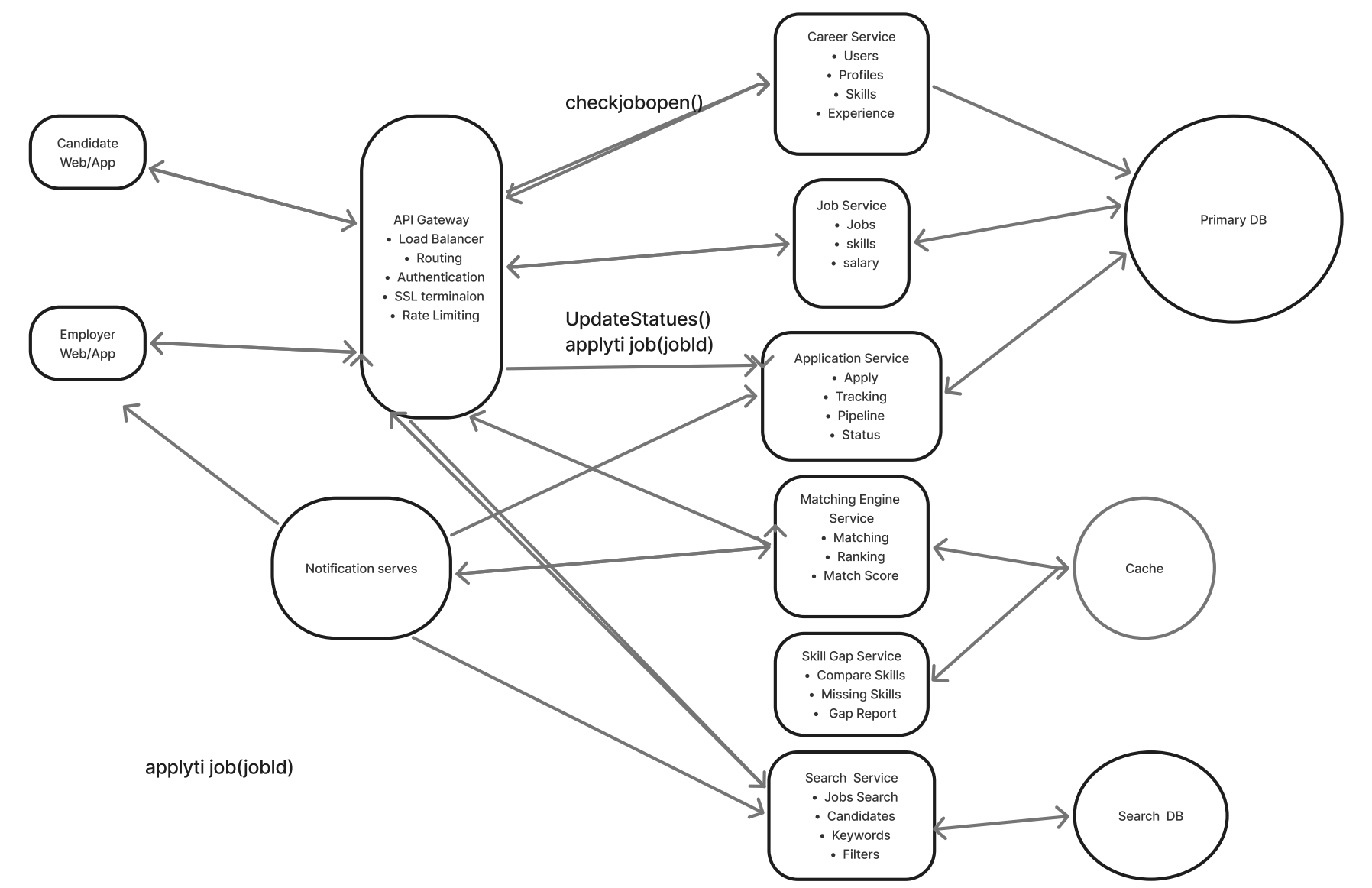
{ name, email , password, role ,skills}

- PATCH/Profile ——→ 200{name, skills ,experience,education, career_goals,portfilo}
- POST/User/login ——→token{

email, password

}

- POST/Jobs ——→jobId{
  title , description , salary , employer_Id, skills}
- Get/ Job ——→ Jobs[]{title , skills , type , salery}
- Post/jobs/ {job id}/ application——→applicationId{candidate_Id}
- Post/jobs/ {job id}/ application——→409{candidate_Id}
- PATCH /applications/{applicationId}/status → 200
  {status:true/False}
- GET /candidates → candidates[]
  { keyword, skill, experience}
- GET /candidates/{candidateId}/recommendations/jobs → jobs[]
- GET /employers/{employerId}/recommendations/candidates → candidates[]
- GET /candidates/{candidateId}/skill-gaps → missingSkills[]
  { jobId}
- GET /notifications → notifications[]
-
  PATCH /notifications/{notificationId}/read → 200
  notificationId


Candidate Web/App
Employer Web/App
API Gateway
Load Balancer
Routing
Authentication
SSL terminaion
Rate Limiting
checkjobopen()
UpdateStatues()
applyti job(jobId)
Career Service
Users
Profiles
Skills
Experience
Job Service
Jobs
skills
salary
Application Service
Apply
Tracking
Pipeline
Status
Matching Engine Service
Matching
Ranking
Match Score
Skill Gap Service
Compare Skills
Missing Skills
Gap Report
Search Service
Jobs Search
Candidates
Keywords
Filters
Notification serves
Primary DB
Cache
Search DB
applyti job(jobId)